

1. Do you want to compare values?

Charts and graphs are perfect for comparing one or many value sets, and they can easily show the low and high values in the data sets. To create a comparison chart, use these types of graphs:

- Column
- Mekko
- Bar
- Pie
- Line
- Scatter Plot
- Bullet

2. Do you want to show the composition of something?

Use this type of chart to show how individual parts make up the whole of something, like the device type used for mobile visitors to your website or total sales broken down by sales rep.

To show composition, use these charts:

- Pie
- Stacked Bar
- Mekko
- Area
- Waterfall

3. Do you want to understand the distribution of your data?

Distribution charts help you to understand outliers, the normal tendency, and the range of information in your values.

Use these charts to show distribution:

- Scatter Plot
- Mekko
- Line
- Column
- Bar

4. Are you interested in analyzing trends in your data set?

If you want to know more information about how a data set performed during a specific time period, there are specific chart types that do extremely well.

You should choose a:

Line
Dual-Axis Line
Column

5. Do you want to better understand the relationship between value sets?

Relationship charts can show how one variable relates to one or many different variables. You could use this to show how something positively affects, has no effect, or negatively affects another variable.

When trying to establish the relationship between things, use these charts:

Scatter Plot
Bubble
Line

How to describe pie charts

The pie chart is primarily used to illustrate how different parts make up a whole. The best way to present your data in a pie chart is to compare the categories with each other. The following comparison words can be used interchangeably:

- to compare
- compared to
- as opposed to
- versus
- more than
- the majority of
- only a small minority
- greater than
- less than

How to describe line graphs

Now, when you know how to describe bar charts, what about line graphs? This type of chart converts information into points on a grid that is connected with a line to represent trends, changes, or relationship between objects, numbers, dates, etc. These lines show movement over time affected by the increase or decrease in the key factors.

To express the movement of the line, you should use appropriate verbs, adjectives, and adverbs depending on the kind of action you need to show. For this, you should use the following vocabulary:

Verbs: rise, increase, grow, go up to, climb, boom, peak, fall, decline, decrease, drop, dip, go down, reduce, level up, remain stable, no change, remain steady, stay constant, stay, maintain the same level, crash, collapse, plunge, plummet.

Adjectives: sharp, rapid, huge, dramatic, substantial, considerable, significant, slight, small, minimal, massive.

Adverbs: dramatically, rapidly, hugely, massively, sharply, steeply, considerably, substantially, significantly, slightly, minimally, markedly.

There is also a list of adverbs to describe the speed of a change: rapidly, quickly, swiftly, suddenly, steadily, gradually, slowly.

Bar Chart:

Represents values across different categories using bars.
Height of each bar represents the value.
Clustered bar charts compare multiple value sets side by side.
Stacked bar charts show the composition of each category.

Line Chart:

Shows trends over time or compares multiple value sets.
Each value set is represented by a line.
Patterns and fluctuations between lines are easily comparable.

Pie Chart:

Typically used for proportions or percentages but can be used for basic comparisons.
Each value set is represented by a slice of the pie.
Sizes of the slices indicate relative magnitudes.

Area Chart:

Similar to a line chart, but the area below the lines is filled.
Enables easier comparison of magnitudes between value sets.
Suitable for comparing cumulative values or proportions over time.

Scatter Plot:

Compares two or more variables to understand their relationship.
Each data point represents a combination of values from the sets.
Observing distribution and patterns in plotted points is possible.

Radar Chart:

Compares multiple variables on different axes.
Each value set is represented by a line or shape.
Lengths or areas of lines/shapes indicate variations and similarities.

Histogram:

Compares the frequency or distribution of values within a single set.
Data is displayed in bins or intervals.
Height of each bar represents the number of occurrences in the range.

Box Plot:

Displays the distribution of a dataset using quartiles.
Shows minimum, maximum, median, and quartiles.
Provides insights into data spread and skewness.

Bubble Chart:

Variation of a scatter plot with data points represented as bubbles.
Size of bubbles represents a third variable.
Allows comparison of three different dimensions of data.

Waterfall Chart:

Tracks cumulative effect of sequentially introduced positive and negative values.
Illustrates financial data or breakdown of changes over time.

Heat Map:

Uses colors to represent values in a matrix or table.
Visualizes patterns, correlations, or density of data.
Effective for comparing data across categories or dimensions.

Gantt Chart:

Bar chart illustrating project schedule or timeline.
Shows start and end dates of tasks or activities.

Visualizes project progress, dependencies, and resource allocation.

Treemap:

Displays hierarchical data using nested rectangles.

Size of each rectangle represents a value.

Visualizes proportions and hierarchies simultaneously.

Polar Chart:

Also known as radar chart with multiple axes.

Compares multiple variables with respect to a central point.

Uses points or lines extending from the center.

Certainly! Here's a brief description of how each graph is represented visually:

Bar Chart:

Categories are represented along the x-axis, and the values are shown using vertical bars of varying heights.

Line Chart:

The x-axis represents time or a continuous variable, while the y-axis represents the values. Lines are drawn to connect data points, showcasing trends or comparisons.

Pie Chart:

A circular chart where each value set is represented by a slice of the pie. The size of each slice corresponds to the proportion or magnitude of the value.

Area Chart:

Similar to a line chart, but the area below the lines is filled with color or pattern, creating shaded regions. The filled areas help compare magnitudes between value sets.

Scatter Plot:

Data points are plotted on a Cartesian coordinate system, with one variable represented on the x-axis and another variable on the y-axis. Each point represents a combination of values from the different sets.

Radar Chart:

Axes radiate from a central point, and each value set is represented by a line or shape extending from the center. The lengths or areas of lines/shapes indicate variations and similarities between variables.

Histogram:

Data is divided into bins or intervals, which are represented on the x-axis. The height of each bar represents the frequency or number of occurrences within that bin.

Box Plot:

A box-and-whisker diagram that displays the minimum, maximum, median, and quartiles of a dataset. It provides a visual representation of the spread and skewness of the values.

Bubble Chart:

Similar to a scatter plot, but each data point is represented by a bubble. The size of each bubble corresponds to a third variable, adding an extra dimension to the comparison.

Waterfall Chart:

Composed of horizontal bars that start at a baseline and go up or down to represent positive or negative values. The cumulative effect of values is shown through the progression of bars.

Heat Map:

A matrix or table where colors are used to represent values. Each cell is filled with a color gradient, with darker or lighter shades indicating higher or lower values.

Gantt Chart:

Consists of horizontal bars that represent tasks or activities. The length of each bar represents the duration, and they are placed along a timeline to show start and end dates.

Treemap:

Hierarchical data is represented by nested rectangles, with each rectangle's size indicating a value. The nesting structure represents the hierarchy.

Polar Chart:

Axes radiate from a central point, and each value set is represented by points or lines extending from the center. The comparison between variables is made by their distances or angles from the center.

Bar Chart Example: Suppose you have sales data for different products in a store. You can create a bar chart to compare the sales of each product.

Example Dataset:

- Product A: 50 units sold
- Product B: 80 units sold
- Product C: 65 units sold

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Line Chart Example: Let's say you want to compare the temperature variations throughout the day in a city.

Example Dataset:

- 8 AM: 20°C
- 12 PM: 25°C
- 4 PM: 23°C
- 8 PM: 18°C

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Pie Chart Example: Consider a survey where respondents are asked about their preferred smartphone brands.

Example Dataset:

- Brand A: 30%
- Brand B: 45%
- Brand C: 25%

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Area Chart Example: Suppose you want to compare the total website visits over a week.

Example Dataset:

- Monday: 200 visits
- Tuesday: 300 visits
- Wednesday: 250 visits
- Thursday: 400 visits
- Friday: 350 visits

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Scatter Plot Example: Let's say you want to compare the relationship between the study hours and exam scores of students.

Example Dataset:

Study Hours (hours)	Exam Score (%)
4	75
6	80
3	65
7	85
5	70

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imgur.com

Radar Chart Example: Suppose you want to compare the performance of different athletes across various sports.

Example Dataset:

Athlete	Basketball	Swimming	Running	Cycling
Athlete A	90	80	70	85
Athlete B	80	90	75	70
Athlete C	70	85	95	80

The image you are requesting does not exist or is no longer available.

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Histogram Example: Let's say you want to compare the distribution of ages in a population.

Example Dataset:

- 20, 25, 30, 22, 40, 50, 45, 35, 28, 32, 42, 38, 27, 29, 33, 37, 24

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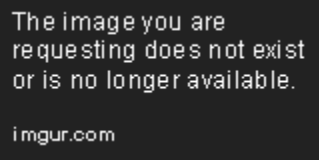
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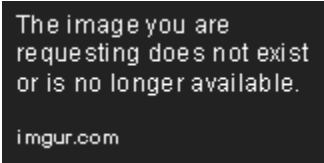


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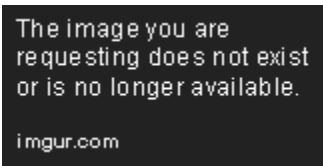
5	70
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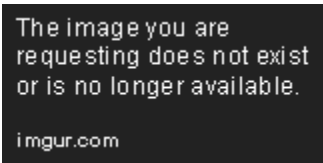
Athlete	Basketball	Swimming	Running	Cycling
Athlete A	90	80	70	85
Athlete B	80	90	75	70
Athlete C	70	85	95	80



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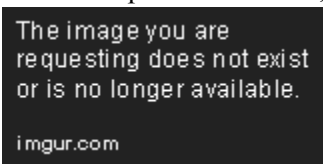
These examples showcase the basic structure and visual representation of each plot. You can customize the appearance, labels, and formatting based on your specific requirements and the tools you are using to create the graphs.

Certainly! Here are examples of some additional types of graphs:

Box Plot Example: Let's say you want to compare the distribution of salaries among employees in different departments of a company.

Example Dataset:

- Department A: \$50,000, \$55,000, \$60,000, \$65,000, \$70,000
- Department B: \$45,000, \$50,000, \$55,000, \$60,000, \$65,000

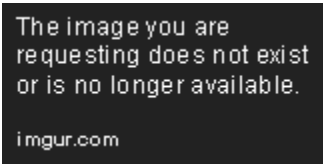


Bubble Chart Example: Suppose you want to compare the sales revenue, advertising expenditure, and market share of different products.

Example Dataset:

Product	Sales Revenue (\$ million)	Advertising Expenditure (\$ million)	Market Share (%)

Product A	10	5	20
Product B	15	8	30
Product C	8	3	15
Product D	12	6	25



Waterfall Chart Example: Let's say you want to compare the monthly financial changes in a company's revenue.
Example Dataset:

Month	Revenue Change (\$)
Jan	+\$100,000
Feb	-\$50,000
Mar	+\$75,000
Apr	-\$40,000
May	+\$60,000

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Heat Map Example: Suppose you want to compare the sales performance of different products across different regions.

Example Dataset:

Region	Product A	Product B	Product C
North	100	80	120
South	90	110	100
East	80	70	85
West	95	100	105

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Gantt Chart Example: Let's say you want to compare the timeline and progress of tasks in a project.

Example Dataset:

Task	Start Date	End Date
Task A	2023-01-01	2023-01-10
Task B	2023-01-05	2023-01-20
Task C	2023-01-15	2023-02-05
Task D	2023-02-01	2023-02-15